

Inrush Current Limiter ICL-150

NTC Thermistor-Based Limiter for High-Reliability Switching Power Supplies

1. Scope

This datasheet defines the Inrush Current Limiter ICL-150, a series of negative-temperature-coefficient (NTC) thermistors developed for the PRM-3A Power Regulation Module Assembly and qualified for use across the MicroCo Power Systems product family. The ICL-150 limits inrush current at power-on by presenting a high cold resistance that self-heats to a low steady-state resistance within the first AC cycle after energization, protecting downstream bridge rectifiers, bulk electrolytic capacitors, and switching FETs from peak transient currents that would otherwise exceed device ratings.

The ICL-150 family is offered in two variants — the standard ICL-150-S (commercial / industrial operating range) and the ICL-150-HT (extended-temperature variant for high-reliability and avionics applications). Mechanical envelope, lead arrangement, and pad layout are identical across variants; differences are limited to the thermistor body composition and the operating temperature range.

2. Electrical Characteristics

Parameter	Symbol	Min	Typ	Max	Unit	Conditions
Cold resistance (25°C)	R _■	13.5	15.0	16.5	Ω	DC measurement, 1 mA test current
Hot resistance (steady-state)	R _h	0.080	0.095	0.115	Ω	I = 8 A continuous, T _{amb} = 25°C
Continuous current rating	I _{ss}	—	10.0	12.0	A	T _{amb} ≤ 50°C, free air
Peak surge current	I _{pk}	—	—	80	A	Single half-cycle, 8.3 ms
Maximum operating voltage	V _{max}	—	—	275	V _{AC}	Continuous
Maximum operating voltage (DC)	V _{max,DC}	—	—	400	V _{DC}	Continuous
Energy rating (single transient)	J _{max}	—	—	180	J	Capacitive load, 25°C, single shot
Thermal time constant	τ	—	22	—	s	From 25°C ambient, no air flow
Dissipation constant	δ	11	—	—	mW/°C	Still air, 25°C
B-value (25/85°C)	B	3400	3550	3700	K	Per IEC 60539

3. Mechanical Specifications

Parameter	Value	Tolerance	Method
Body diameter	20.0 mm	±0.5 mm	Caliper, post-glaze
Body thickness	6.5 mm	±0.3 mm	Caliper, axial
Lead spacing (centerline)	10.0 mm	±0.3 mm	Optical comparator
Lead length (untrimmed)	30.0 mm	±2.0 mm	Caliper
Lead diameter	0.81 mm	±0.03 mm	Micrometer
Lead material	Tin-plated copper	—	Per IPC-A-610 G
Mass (per unit)	8.4 g	±0.4 g	Analytical balance
Mounting	Through-hole, 1.1 mm finished hole	—	Per PCB drawing 5018-A
Marking	Laser-etched: part number, lot code, date-code		Visual, 5× magnification

4. Performance Characteristics

4.1 Inrush Limiting Effectiveness. When energized into a representative capacitive load (820 µF bulk electrolytic, 16 mΩ ESR) at 230 V_{AC} mains, peak inrush current is limited to ≤ 14.2 A at 25°C ambient, against an unprotected peak of ≈ 84 A. Effectiveness is preserved across the operating temperature range with ≤ 8% degradation at +85°C.

4.2 Steady-State Behavior. At nominal continuous current (8 A), the device self-heats to a stable surface temperature of approximately 115°C above ambient, presenting a steady-state resistance of 0.095 Ω. Continuous-operation power dissipation: ≈ 6.1 W at 8 A.

4.3 Recovery After Brief Interruption. Following a brief AC mains interruption (≤ 100 ms), the device retains residual self-heating and presents a reduced cold resistance on re-energization, producing a second-strike inrush current of ≤ 32 A within 2 s of the original power-on event. Full cold-state recovery (return to R_{DS(on)} ≥ 14.0 Ω) requires ≥ 90 s of cooling at 25°C still air. Applications requiring repeated cycling within the recovery window should refer to §5.2.

4.4 Cycle Endurance. Qualified for ≥ 100,000 power-on / power-off cycles at rated load with no measurable degradation in R_{DS(on)}, R_h, or inrush limiting performance. See §5.3.

5. Qualification Data

Three (3) production lots (n = 30 units / lot) were subjected to the full qualification test sequence. All 90 units passed all acceptance criteria with no parametric drift outside the specification limits of §2 and §3. Detailed test reports are archived under QR-ICL150-001 through QR-ICL150-090.

Parameter	Specification	Standard
Operating temperature (ICL-150-S)	–25°C to +85°C	IEC 60068-2-1 / -2
Operating temperature (ICL-150-HT)	–55°C to +125°C	MIL-STD-810H Method 501.7 / 502.7
Storage temperature	–55°C to +150°C	IEC 60068-2-1 / -2
Thermal cycling	1,000 cycles (–40°C to +85°C, 30 min dwell)	JESD22-A104D Condition G
Humidity (operating)	5%–95% RH, non-condensing	MIL-STD-810H Method 507.6
Vibration	20–2000 Hz, 14.6 G _{rms} , 3 axes × 1 h	MIL-STD-810H Method 514.8 Cat. 24
Mechanical shock	1,500 G, 0.5 ms half-sine, 6 directions	MIL-STD-810H Method 516.8
Solderability	245°C ± 5°C, 5 s, dipped	IPC J-STD-002 Test A
Resistance to soldering heat	260°C, 10 s, single dip	IEC 60068-2-20 Test Tb
Cycle endurance	100,000 cycles at I _{ss} = 8 A	MicroCo qualification plan QP-ICL150
Service life	20 years (continuous operation)	MicroCo qualification plan QP-ICL150

6. Compliance and Standards References

Standard	Title	Application
IEC 60539-1	Directly heated negative temperature coefficient thermistors	§2
IEC 60068-2	Environmental testing	§5
MIL-STD-810H	Environmental Engineering Considerations and Laboratory Tests	§5
JESD22-A104D	Temperature Cycling	§5
IPC J-STD-002	Solderability Tests for Component Leads, Terminations, Lugs	§5
UL 1434	Standard for Thermistor-type Devices	§7
IPC-A-610 G	Acceptability of Electronic Assemblies	§3
RoHS 2011/65/EU	Restriction of Hazardous Substances Directive	§7
REACH (EC) 1907/2006	Registration, Evaluation, Authorisation and Restriction of Chemicals	§7

7. Compliance Declarations

RoHS: Compliant with Directive 2011/65/EU, including the (EU) 2015/863 amendment (Annex II revision adding four phthalates). No lead in solder, thermistor body, or terminations. Tin-plated copper leads use a matte-tin finish ($\geq 8 \mu\text{m}$) qualified against tin-whisker growth per JEDEC JESD201A.

REACH: No SVHCs above 0.1% w/w per the candidate list dated 2026-01-23. Full material declaration available on request.

UL Recognition: Recognized under UL File E523114, component category XGPU2 (Thermistor-type Devices), with Conditions of Acceptability documented under file cover sheet rev. 4 (2026-02-12).

Conflict Minerals: Sourced in conformance with Section 1502 of the U.S. Dodd-Frank Act. CMRT (Conflict Minerals Reporting Template) v6.31 available on request.

8. Ordering Information

Part Number	Variant	Operating Range	Packaging
ICL-150-S	Standard (commercial / industrial)	-25°C to +85°C	Bulk, 500 / box
ICL-150-S-T	Standard, tape & reel	-25°C to +85°C	Reel, 1,000 / reel
ICL-150-HT	Extended temperature (avionics / military)	-55°C to +125°C	Bulk, 500 / box, sealed bag
ICL-150-HT-T	Extended temperature, tape & reel	-55°C to +125°C	Reel, 1,000 / reel

9. Revision History

Rev.	Date	Author	Description
A	2026-03-13	K. Møller	Initial release

Approved By	Title	Signature	Date
K. Møller	Component Engineer	// signature on file //	2026-03-13
R. Patel	Quality Engineering Manager	// signature on file //	2026-03-13
L. Tanaka	Director, Hardware Engineering	// signature on file //	2026-03-14